Sample Public School

Unit Test: Relational Databases and SQL

Subject: Computer Science Class: Class 12

Time allowed: 45 minutes **Maximum marks: 20**

TEACHER'S COPY · ANSWER KEY AT THE END

General instructions

1. All questions are compulsory.
2. Write SQL keywords in capital letters.
3. Section C questions carry 4 marks each.

SECTION A

Multiple Choice Questions

Choose the correct option. (2 × 1 = 2 marks)

1. Which key uniquely identifies each row of a table? [1]

A) foreign key B) primary key

C) composite attribute D) alternate index

2. Which SQL command removes a table's structure and all its data permanently? [1]

A) DELETE B) TRUNCATE

C) DROP D) REMOVE

SECTION B

Short Answer Questions

Answer briefly. (2 × 3 = 6 marks)

3. What is a foreign key? Explain with an example. [3]

4. What is normalisation? Name the three anomalies it removes. [3]

SECTION C

Long Answer Questions

Answer in detail. (3 × 4 = 12 marks)

5. Differentiate between DDL and DML, giving two examples of commands for each. [4]

6. Write an SQL query to display the name and marks of students who scored more than 80, sorted by marks from highest to lowest, from the table STUDENT(roll_no, name, marks). [4]

7. Explain 1NF, 2NF and 3NF briefly. [4]

— End of question paper —

Answer key and marking scheme

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Section A · Multiple choice

1. 1 mark · Moderate

(B) primary key

2. 1 mark · Moderate

(C) DROP

Section B · Short answer

3. 3 marks · Moderate

- A foreign key is a column in one table that refers to the primary key of another table and links the two tables (2 marks).
- A correct example, e.g. customer_id in ORDERS referring to CUSTOMERS(customer_id) (1 mark).

4. 3 marks · Moderate

- Normalisation is organising data into well-structured tables to reduce redundancy and dependency (1.5 marks).
- It removes the insertion, update and deletion anomalies (1.5 marks, 0.5 each).

Section C · Long answer

5. 4 marks · Moderate

- DDL (Data Definition Language) defines or changes the structure of database objects (1 mark); examples CREATE, ALTER, DROP (1 mark).
- DML (Data Manipulation Language) works on the data stored in tables (1 mark); examples INSERT, UPDATE, DELETE (1 mark).

6. 4 marks · Moderate

SELECT name, marks FROM STUDENT WHERE marks > 80 ORDER BY marks DESC;
- SELECT name, marks (1 mark), FROM STUDENT (1 mark), WHERE marks > 80 (1 mark), ORDER BY marks DESC (1 mark).

7. 4 marks · Moderate

- 1NF: every column holds atomic values and there are no repeating groups (1 mark).
- 2NF: in 1NF and no partial dependency of a non-key attribute on part of a composite key (1.5 marks).
- 3NF: in 2NF and no transitive dependency; non-key attributes depend only on the key (1.5 marks).